3. Simple Segmentation

> *An image contains objects clearly distinguishable from the background based on intensity.*

(a) Interpretation of the scenario and segmentation requirements

When objects in an image have intensity values that are clearly different from the background - typically producing a bimodal histogram with two distinct peaks (one for the object, one for the background) - the segmentation task is to separate these two regions into a binary result: foreground (object) and background. The requirement is a method that reliably identifies the intensity value separating the two populations.

(b) Key features enabling separation

• A bimodal (or near-bimodal) histogram with two clearly separated peaks.

• A visible valley (low-frequency region) between the two peaks, indicating a natural dividing intensity value.

• Relatively uniform intensity within the object and within the background (low internal noise/variation).

• Minimal overlap between the object and background intensity ranges.

(c) Applying a suitable segmentation technique

Global thresholding is applied: a single threshold value T is chosen (e.g. at the valley between the two histogram peaks, which can be found automatically using Otsu's method), and every pixel is classified as:

$g(x, y) = 1$ (object) if $f(x, y) >= T$, $g(x, y) = 0$ (background) if $f(x, y) < T$

(d) Justification

Because the object and background intensities are already well separated and each region is fairly uniform, a simple global threshold is sufficient to achieve accurate segmentation. More complex techniques such as region growing, split-and-merge, or edge-based segmentation are unnecessary here since they are designed to handle more complicated cases (uneven illumination, overlapping intensities, or textured regions). Global thresholding is computationally efficient, easy to implement, and gives an accurate result precisely because the underlying condition - a clear intensity gap - is already satisfied.

(e) Summary

The clear intensity separation between object and background is the key feature enabling segmentation, and a global threshold placed at the histogram valley between the two peaks provides a simple, efficient, and accurate binary segmentation for this scenario.